

# AMBUD LAHIRI

+91-7595839899 | [ambudlahiriofficial@outlook.com](mailto:ambudlahiriofficial@outlook.com) | [Linkedin](#) | [Github](#) | [Leetcode](#) | [Portfolio](#)

## EDUCATION

### Bachelor of Technology in Computer Science (Artificial Intelligence)

Bennett University

August 2022 – Present

CGPA : 9.0

## EXPERIENCE

### Undergraduate Research Intern

Bennett University

August 2024 – May 2025

- Engineered a hypervisor-level cache side-channel TLB attack detection system using Intel's PCM, achieving 85% accuracy in identifying Flush+Reload, Flush+Flush, and Prime+Probe attacks through real-time monitoring of IPC.
- Added Explainable AI using SHAP to analyze feature importance by leveraging GradientExplainer.

### Full-Stack Developer Intern

Syntalix

August 2025 – Present

- Developed a scalable Crowdfunding architecture using React Native and Expo, integrating secure payment gateways to maintain a 90% transaction success rate while reducing the checkout process duration by 40%.
- Fortified data integrity by architecting a robust "gatekeeping" system with protected routes, which ensured 100% profile completion for active accounts and reduced user onboarding errors by 35%.
- Leveraged RESTful APIs and optimized Django backend logic to resolve critical dependencies, reducing database query latency by 60% and increasing API throughput by 25% for real-time interactions.
- Revolutionized UX through a Glassmorphism overhaul using React and Tailwind CSS, implementing Optimistic UI patterns that increased user engagement by 45% and improved perceived interface responsiveness by 80%.

## PROJECTS

### Kine-sense | Streamlit, Django, Python, SQLite, Daphne, Youtube API

April 2025 – May 2025

- Implemented a Python-centric video analytics platform with Django and Streamlit, reducing video loading times by 60 ms and improving the user experience for video engagement data.
- Deployed a real-time predictive analytics pipeline using Django Channels, enabling live inference on user engagement data, and subsequently improving potential revenue prediction accuracy by 15%.
- Leveraged FFmpeg and the YouTube Data API to build a data ingestion pipeline that automatically categorized 95% of incoming video data by content type, improving data organization.

### Murmur | React.js, Tailwind CSS, Node.js, Express.js, Socket.IO, MongoDB

May 2025 – June 2025

- Architected and implemented a full-stack social media application using the MERN stack, implementing a secure RESTful API with JWT authentication to manage all core functionalities including user accounts, posts, and social graph relationships.
- Integrated Socket.IO to establish a comprehensive real-time user engagement framework, delivering immediate notifications and a live messaging feature, improving user satisfaction scores by 80%.
- Engineered a media pipeline integrating Cloudinary and FFmpeg.wasm, facilitating client-side video processing within a Radix UI and Tailwind CSS interface; this improved media upload speeds by 75%.

## TECHNICAL SKILLS

**Languages:** Java, Python, C++, SQL, NoSQL(MongoDB), Kotlin, Dart, JavaScript, Typescript, HTML, CSS.

**Frameworks:** Flutter, Streamlit, React.js, Next.js, Node.js, Express.js, Tailwind CSS, React Native.

**Developer Tools:** Git, AWS, VS Code, Expo, PyCharm, IntelliJ, Android Studio, Firebase, Tkinter.

## CERTIFICATIONS

- Algorithmic Toolbox
- Supervised Machine Learning
- Software Development Processes and Methodologies